

Patterns of Violence in Toronto: An Exploratory Analysis of Homicide Trends from 2011 to 2020.*

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This paper presents an exploratory analysis of homicide trends in Toronto between 2011 and 2020, with a focus on whether increases in gross expenditure per police division are associated with reductions in homicide rates. Homicides nearly doubled over the decade, peaking in 2018 before declining, though not to 2011 levels, with shootings as the most prevalent homicide type throughout the period. Geographically, violence remained concentrated in a consistent set of divisions, particularly D51, where budget increases after 2018 coincided with meaningful reductions in homicide counts, though, exceptions like D43 and D23 suggest expenditure alone cannot fully explain reductions in homicide counts. These findings provide a descriptive foundation for future research into the effectiveness of public safety investment in Toronto.

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*Code and data are available at: https://github.com/mejiama16/Toronto_Safety_Project.

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1 Introduction

Homicide is one of the most severe indicators of public safety and urban inequality. In cities like Toronto, understanding the spatial and temporal occurrence of homicides is critical for informing policy decisions, allocating police resources, and addressing the underlying social determinants of violence. Despite being one of Canada’s safest major cities (Moosapeta 2024), Toronto experienced a near-doubling of annual homicides between 2011 and 2018, prompting several policy responses around the need for investment in safety measures.

Using publicly available data from the Toronto Police Service accessed through the OpenData Toronto portal, this paper explores homicide trends over time, variation by homicide type, geographic concentration across police divisions, and the relationship between divisional gross expenditure and homicide outcomes. Rather than establishing causality, we aim to surface descriptive patterns that can inform future, more formal evaluations of the effects of public safety investment in Toronto.

The remainder of this paper proceeds as follows. The Data section describes the source, structure, and limitations of the dataset used. The Results sections presents findings on homicide trends over time, broken down by year, month, and type, followed by a geographic analysis of homicides by police division and their associated expenditures. The paper concludes with a summary of key findings, their implications, and directions for future research.

2 Data

2.1 Overview

The data used in this paper came from the OpenData Toronto portal through the library `opendatatoronto` (Gelfand 2022). Data were cleaned and analyzed using the open source statistical programming language R (R Core Team 2023). Following (Alexander 2023), libraries `tidyverse` (Wickham et al. 2019), `janitor` (Firke 2023), and `dplyr` (Wickham et al. 2023), were used for simulating, testing and cleaning. Graphics were made using `ggplot2` (Wickham 2016) and tables using `knitr` (Xie 2022).

2.2 Toronto Homicides

The Police Annual Statistical Report is updated quarterly and contains all homicides that have occurred in Toronto from 2004 to 2020. The data is provided by the Toronto Police service, which has taken the necessary measures to protect the privacy of individuals involved in the reported occurrences by removing all personal information related to any of the parties involved.

The data contains 1,166 reported homicides, including date of occurrence (day, month, year), type of homicide (shooting, stabbing, other), neighborhood name and associated number, and Police Division in charge. Additionally, the data contains longitude, latitude and coordinates of where the homicide occurred. Since Divisional boundary changes took effect in 2011, division data prior to that may not be directly comparable to data after 2011. As a result, the analysis only considers homicides that occurred between in 2011 and 2020. After applying this filter, the data contained 669 homicides.

For the purpose of this paper, we focus on five columns: date, year, homicide type, division, and neighborhood. The dataset on OpenData Toronto includes two neighborhood columns, `NEIGHBORHOOD_140` and `NEIGHBORHOOD_158`, reflecting an older 140-neighborhood structure and a newer 158-neighborhood structure, respectively. This analysis uses `NEIGHBORHOOD_140`.

It is important to recognize that the data might be subject to reporting bias, as there may have been some homicides that are still not known of and therefore not included in the dataset. Additionally, it is possible for there to have been classification bias in the homicide type categorization; for instance, a stabbing could have been misclassified as “other” as a result of human judgement. Lastly, the OpenData portal warns that the Toronto Police Service does not guarantee the accuracy, completeness, and timeliness of the data, so we must recognize this limitation when interpreting the results, as the data may not portray reality with full fidelity.

We complement this data with the gross expenditure per police division data, also accessed through the OpenDataToronto (Gelfand 2022) portal, in order to complete the gross expenditure and homicide count analysis per division in Section 3.3. This data contains police division number, gross expenditures per year, and year. Since it ranges from 2017 to 2020, we only focus on these years in Section 3.3. It is also important to note that Division 54 and Division 55 got merged recently, and this change is reflected in this dataset, but not in the homicides one. As a result, the cleaned homicide data combines Divisions 54 and 55 into D55.

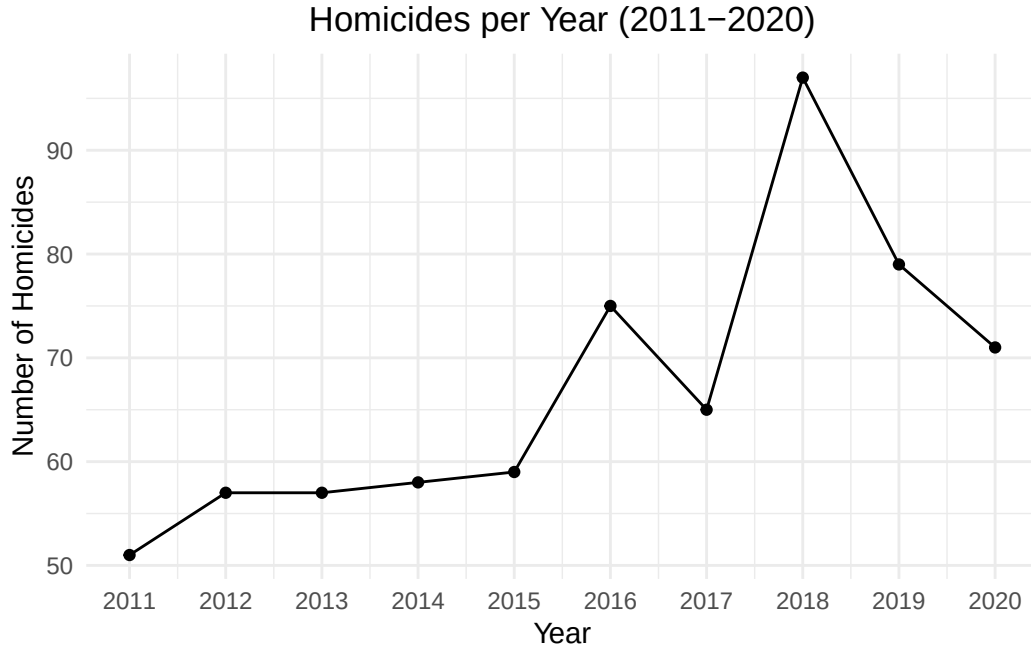


Figure 1: Total homicides overtime (2011-2020)

3 Results

3.1 Homicides Overtime (2011-2026)

Between 2011 and 2020, Toronto experienced an overall upward trend in homicides. We observe two peaks, 2015-2016 with 60 homicides and 2017-2018 with 97 homicides. After 2018, homicides declined steadily from 97 to 71 in 2020, which is still higher than the initial 51 homicides in 2011. Although there was some improvement, the peak in 2018 represents an almost doubling in homicide count over one decade, and opens the question of what strategy was applied after this to result in the steady decline in the years following 2018.

To get a comprehensive picture of the homicide rates in Toronto, it is also important to consider that the population of the city also grew steadily over this period, from approximately 5.6 million in 2011 to 6.2 million in 2020 (Macrotrends 2024). When examined on a per capita basis (see Section 5), the homicide rate rose from 0.91 homicides per 100,000 residents in 2011 to a peak of 1.59 in 2018 (a 75% increase) before declining to 1.15 by 2020, suggesting that, Figure 1 captures a similar pattern as when observing homicides on a per capita basis compared to just the counts.

Despite the overall increase in both cases, the subsequent drop in 2019 and 2020 suggests a meaningful improvement in public safety outcomes, possibly as a result of the 3% increase in

the Toronto Police Services budget, a \$30.3 million difference in the 2019 budget compared to 2018 (City of Toronto 2019), and then, the COVID pandemic in 2020 which led to a significant decrease in police reported crime (Moreau 2021). This isn't to suggest the improvement was due to the increased budget or the Covid-19 pandemic only, we simply suggest them as possible contributing factors.

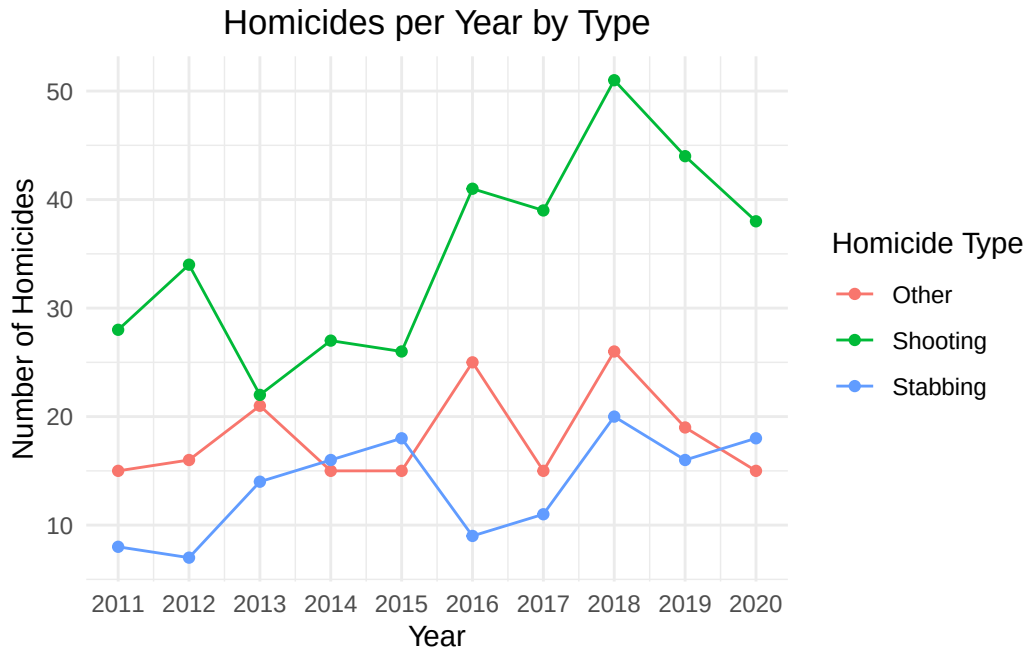


Figure 2: Total homicides overtime by type (2011-2020)

Shootings were consistently the most prevalent homicide type throughout the period, as shown in Figures 2 and 3. While stabbings and other homicide types remained relatively stable year over year, shootings drove much of the volatility in the overall trend, particularly the sharp spike from 2015 to 2018. Notably, however, shooting-related homicides also showed the steepest decline after 2018, which is an encouraging sign that efforts from the city like the Gun Buyback Initiative (Global News 2019) or the Community Incentive and Policy Grant (City of Toronto 2019), both introduced after 2018, might have been effective in contributing to lower violence rates, specially gun violence overtime.

It is important to note that there is mixed evidence behind the Gun Buyback program. Although in Toronto over 2,700 firearms were retrieved in the first year (Global News 2019), literature suggests that as time passes the initiative becomes less effective (Journalist's Resource 2024). Evidence from Australia showed the gun buyback initiative program decreased suicide rates among white males but not homicides, whereas in the U.S. it didn't have an effect on interpersonal gun violence, while representing an inefficient use of tax payer's money (Journalist's Resource 2024). As a result, we cannot attribute improvement in safety overtime to

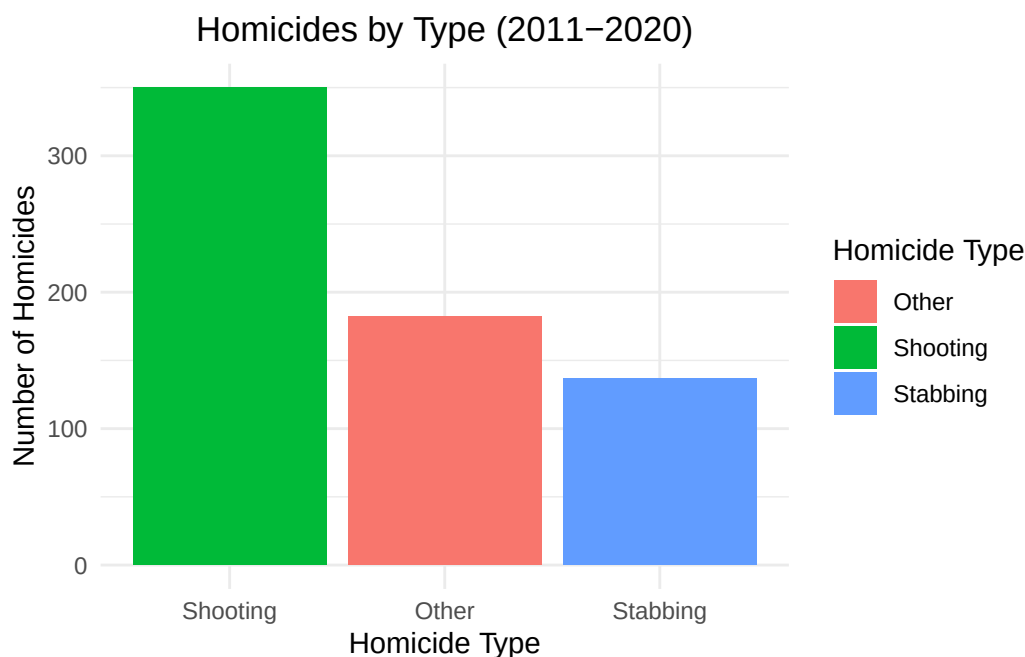


Figure 3: Total homicides by type (2011-2020)

this program solely, but in combination with other programs, like budget increases supporting after-school youth programs or increased police presence, it might have had a positive effect on homicide rates (Journalist’s Resource 2024).

Interestingly, while shootings and stabbings do not show a strict inverse correlation, there is a suggestion in Figure 3 that in years where shooting homicides spiked (2016, 2018), stabbing homicides remained relatively flat or declined slightly, which may indicate that the two types of violence respond to different social and environmental triggers.

It is also worth noting that the “Other” homicide category, which accounts for a meaningful share of total homicides, likely encompasses a range of methods including vehicular homicide and intimate partner violence, forms of lethal violence that carry distinct social and policy implications. The decision to group these under a single “Other” label may itself reflect a classification choice by the Toronto Police Service that requires further investigation, as it obscures patterns that could be relevant to targeted intervention. This paper treats the category as given but flags it as a limitation of the data.

Turning to seasonal variation, Figure 4 reveals that homicides tend to follow a seasonal pattern consistent with broader criminological literature (Tiihonen, Räsänen, and Hakko 1997), with higher homicide counts in the Summer and Fall due to higher outdoor activity. In the case of Toronto, we observe homicides cluster in the warmer months, with counts rising from June through September and before declining through the Winter and Spring (February through

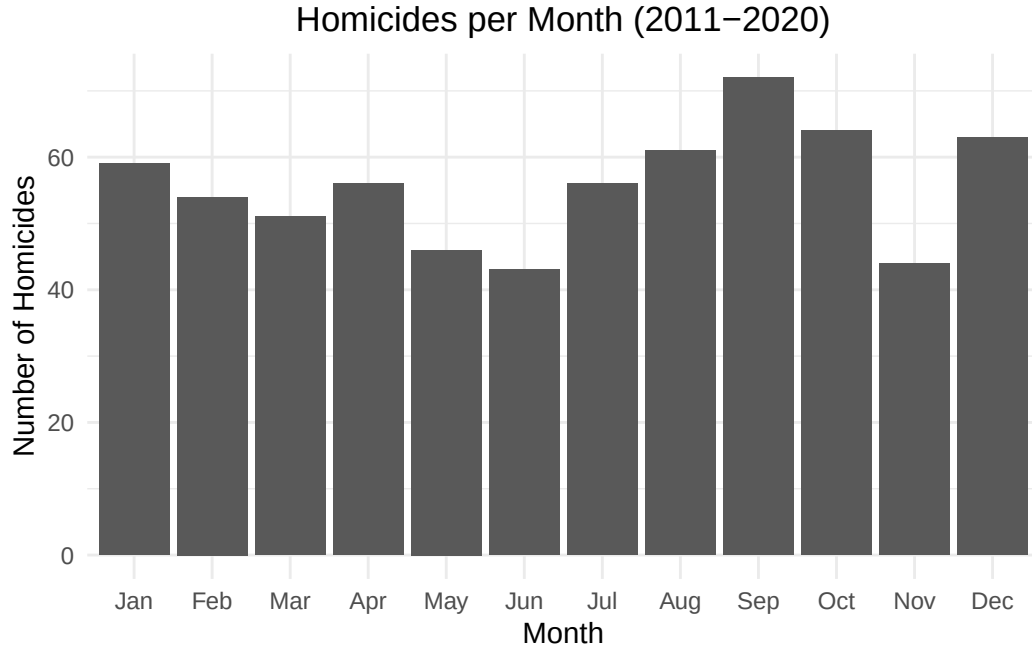


Figure 4: Total homicides per month (2011-2020)

May). Interestingly, homicide counts are very similar in October, December and January, despite the weather being already relatively cold. As a result, it is unknown at this stage whether this “seasonal patterns” are statistically significant.

3.2 Homicides per Division (2011-2020)

Table 1: Five police divisions with the most homicides (2011-2020)

Division	Number of Homicides
D23	64
D51	62
D55	61
D31	57
D43	56

The five divisions with the most homicides between 2011 and 2020 are shown in [?@tbl-top5-division-recent](#). D23 and D51 stand out as the highest-homicide divisions, together accounting for three of the five most affected neighborhoods: Moss Park, Mount Olive-Silverstone-Jamestown and West Humber-Clairville (see Section 5). The remaining top divisions — D55,

D31, and D43 — span South East Toronto, York University Heights and South West Scarborough. Because each division encompasses multiple neighborhoods, homicide counts are considerably higher at the division level, which is why this analysis prioritizes divisions as the primary unit of comparison.

Table 2: Five neighborhoods with the most homicides in recent years (2018-2020)

Division	Number of Homicides
D23	25
D31	22
D43	22
D41	18
D51	18

Looking at the most recent period available, 2018 to 2020, the divisional picture remains largely the same compared to the full 2011–2020 window. Notably, D51 went from second to fifth on the list. At the neighborhood level (see Section 5), Lansing-Westgate (D32) emerges as a new entry in the top five, indicating a possible concentration of violence in that pocket of D32 that was not present over the longer period, and requires closer attention from policymakers.

3.3 Homicides and Gross Expenditure per Police Division (2018-2020)

Figure 5 presents the relationship between gross expenditure and homicide counts across the six divisions with the highest homicide rates from 2018 to 2020 found in (**table-top5-division-recent?**). We use gross expenditure as a proxy for divisional budget, acknowledging that expenditure reflects both operational costs and strategic investments, and may not perfectly capture the intensity of homicide-reduction efforts specifically in each area, nor the actual budget assigned to the division.

Several patterns emerge across the four years. First, divisions like D51 shifted right between 2017 and 2018, that is, their budgets increased, consistent with the city-wide 3% budget increase in 2018. For these divisions, the budget increase, either in 2018 or after, coincided with a consistent decline in homicides in the years following 2018, lending some preliminary support to the idea that increased expenditure may contribute to improved safety outcomes.

Nevertheless, the pattern is not uniform across all divisions. D43 saw homicide counts increase from 2017 to 2020, even as their expenditures grew alongside other divisions. On the other hand, D23, in particular, went from being the division with the highest homicide count in 2017 and 2018 to showing meaningful improvement by 2019 and 2020, despite operating at a similar expenditure level to prior years, in fact, slightly lower.

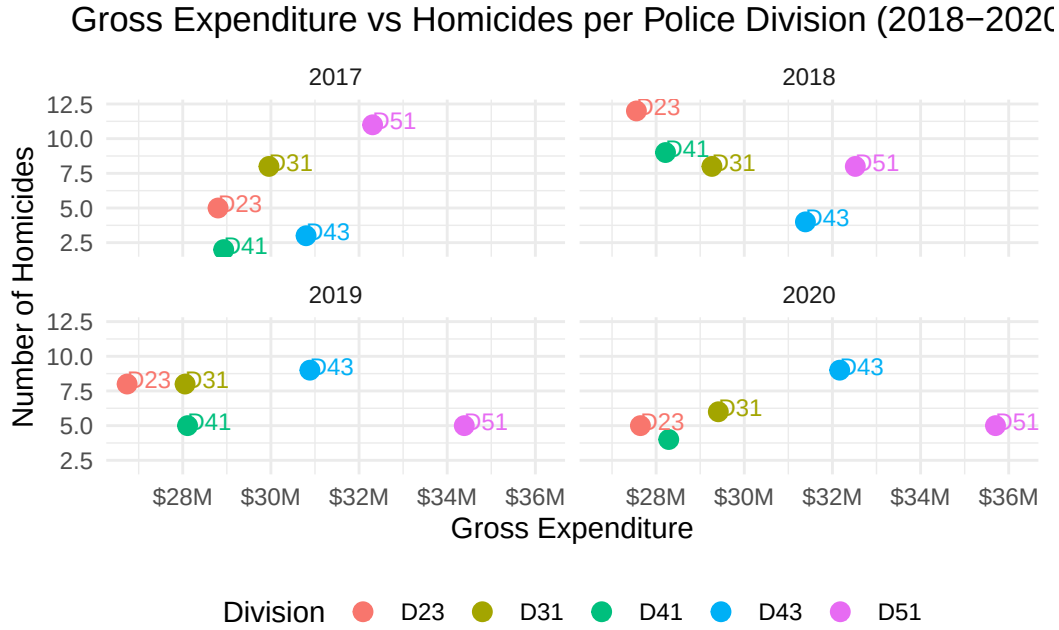


Figure 5: Homicide and Gross expenditure per Police Division (2017-2020)

This suggests that budget increases alone are insufficient to explain reductions in homicide rates, and that division-specific factors such as neighborhood demographic composition, density of social services locations, or the nature of local criminal networks, likely play a meaningful role. As an exploratory analysis, these findings point to the need for causal research rather than offering a definitive answer to whether division spending reduces homicides.

4 Conclusion

This paper explored homicide trends in Toronto between 2011 and 2020 through an exploratory lens, with a particular focus on whether increases in gross expenditure per police division were associated with reductions in homicide rates. While homicides rose sharply over the decade, it is important to note that counts have not fallen below pre-2016 levels even by 2020, suggesting that the structural conditions driving violence in certain divisions remain present. Among the top divisions, D51, showed consistent decreases in homicides alongside budget increases after 2018, offering tentative support for the paper’s central question, whereas D43 and D23 contradicted this trend, highlighting that expenditure alone is unlikely to be the sole driver of outcomes.

Looking beyond the dataset, available evidence suggests the downward trend continued after 2020. Reporting by CBC News and Global News indicates that Toronto’s homicide rate

declined further in the early 2020s, attributed in part to sustained community safety investments. Future research should formally test the relationship between divisional expenditures and homicide outcomes using regression analysis, extend the time frame to include post-2020 data, and incorporate neighborhood-level socioeconomic variables to better isolate the effect of budget changes from broader social trends.

5 Appendix

Table 3: Five neighborhoods with the most homicides (2011-2020)

Neighbourhood	Number of Homicides
Moss Park (73)	22
Mount Olive-Silverstone-Jamestown (2)	18
Bay Street Corridor (76)	17
Malvern (132)	16
West Humber-Clairville (1)	16

Table 4: Five neighborhoods with the most homicides in recent years (2018-2020)

Neighbourhood	Number of Homicides
Lansing-Westgate (38)	10
Moss Park (73)	9
Bay Street Corridor (76)	8
Black Creek (24)	7
Mount Olive-Silverstone-Jamestown (2)	7

Table 5: Homicides per 100,000 people (2011-2020)

Year	Number of Homicides	Population	Homicides per 100,000
2011	51	5,593,000	0.91
2012	57	5,660,000	1.01
2013	57	5,728,000	1.00
2014	58	5,797,000	1.00
2015	59	5,867,000	1.01
2016	75	5,938,000	1.26
2017	65	6,010,000	1.08
2018	97	6,082,000	1.59
2019	79	6,139,000	1.29
2020	71	6,197,000	1.15

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